

Name: _____

Solutions

Date: _____



Year 9 Mathematics

Investigation 3, 2015

Topic – Area and Volume of Similar Figures

In class component

Total Time: **60 minutes**

Reading Time: **5 minutes**

Working Time: **55 minutes**

Equipment: *Scientific Calculator, Take Home Component*

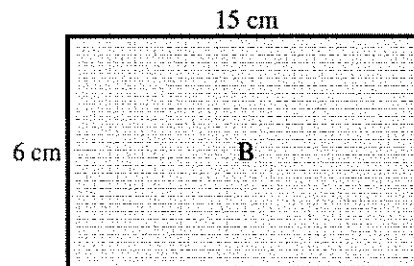
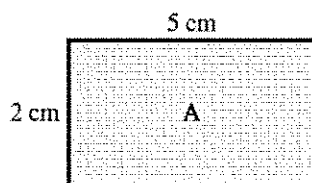
Bonus Marks for Take Home Component: **+ 1 – ready by due date**

+ 1 – reasonable attempt made

AIM: *In this assessment, you will be investigating the area and volume of similar figures.*

1. [5 marks: 1, 1, 1, 1, 1]

Consider the shapes below:



a) Calculate the perimeter of shape A.

14 ✓

b) Calculate the perimeter of shape B.

42 ✓

c) $\frac{\text{Perimeter of B}}{\text{Perimeter of A}} = \frac{42}{14} = 3 \quad \checkmark$

d) By what scale factor were the side lengths of A increased to get B?

3 ✓

e) Explain the relationship between the answers to part (c) and (d).

*WHEN PERIMETER IS INCREASED BY A SCALE FACTOR OF 3
SO ARE THE SIDES OF THE SHAPE. ✓*

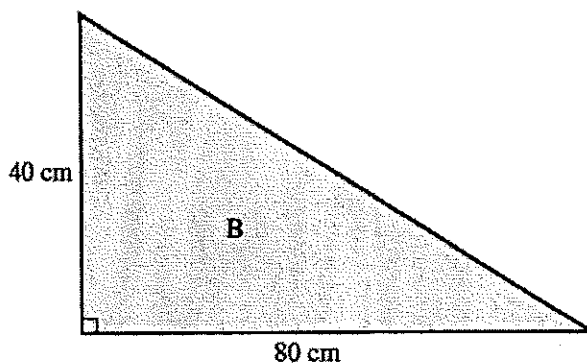
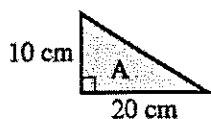
2. [1 mark]

Complete the following sentence:

If the side lengths of any figure are increased by a scale factor of n , then the perimeter of similar figures are increased by a scale factor of n .

3. [5 marks: 1, 1, 1, 1, 1]

Consider the shapes below:



a) Calculate the area of shape A.

100 ✓

b) Calculate the area of shape B.

1600 ✓

c) $\frac{\text{Area of B}}{\text{Area of A}} = \frac{1600}{100} = 16$ ✓

d) By what scale factor were the side lengths of A increased to get B?

4 ✓

e) Explain the relationship between the answers to part (c) and (d).

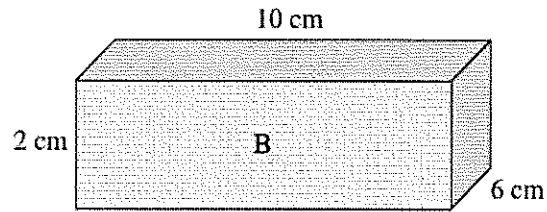
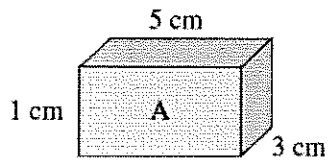
$4^2 = 16$ SCALE FACTOR OF AREA IS SCALE FACTOR OF SIDE LENGTH SQUARED. ✓

4. [1 mark]

Complete the following sentence:

If the side lengths of any figure are increased by a scale factor of n , then the area of similar figures are increased by a scale factor of n^2 .

5. [5 marks: 1, 1, 1, 1, 1]
Consider the shapes below:



- a) Calculate the volume of shape A.

15 ✓

- b) Calculate the volume of shape B.

120 ✓

c) $\frac{\text{Volume of B}}{\text{Volume of A}} = \frac{120}{15} = 8$ ✓

- d) By what scale factor were the side lengths of A increased to get B?

2 ✓

- e) Explain the relationship between the answers to part (c) and (d).

$2^3 = 8$ SCALE FACTOR OF VOLUME IS SCALE FACTOR OF SIDE LENGTH POWERED BY 3.

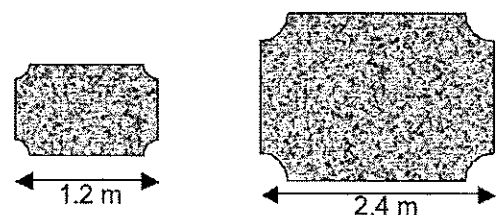
6. [1 mark]
Complete the following sentence:

If the side lengths of any figure are increased by a scale factor of n , then the volume of similar figures are increased by a scale factor of n^3 .

7. [2 marks]
Two kitchen worktops are similar in shape. The perimeter of the smaller worktop is 8 m.

Calculate the perimeter of the larger worktop.

16 m ✓ SCALE FACTOR IS 2. ✓



8. [3 marks]

Two flags are mathematically similar. The larger flag has an area of 7680 cm^2 . Find the area of the smaller flag.

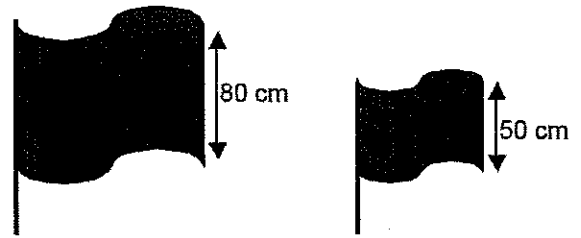
SCALE FACTOR is $\frac{80}{50} = 1.6$ ✓

$7680 \div 1.6^2 = 3000 \text{ cm}^2$ ✓

OR

$1.6 \times 1.6 = 2.56$ ✓

$7680 \div 2.56 = 3000$ ✓



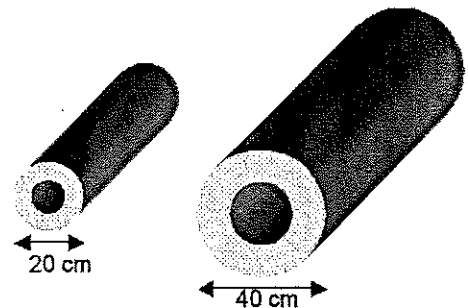
9. [3 marks]

Two metal pipes are similar in shape. The smaller pipe has a volume of 8000 cm^3 . Find the volume of the larger pipe.

SCALE FACTOR is $\frac{40}{20} = 2$ ✓

$2^3 = 8$ ✓

$8000 \div 8 = 1000$ ✓



10. [3 marks]

a) A triangle has perimeter 10 cm and area 4 cm^2 .

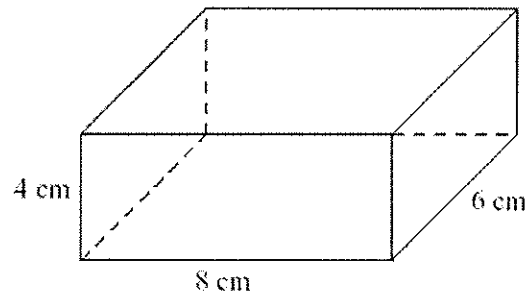
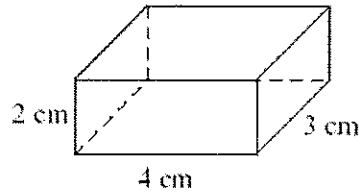
Could you make a similar figure with perimeter 30 cm and area 12 cm^2 ? Why or why not?

No, IF PERIMETER INCREASES BY A SCALE FACTOR OF 3 THEN AREA SHOULD INCREASE BY A SCALE FACTOR OF 9

$4 \times 9 = 36$ NOT 12 cm^2 ✓

11. [7 marks: 1, 2, 2, 1, 1]

Consider the similar figures below:



- a) By what scale factor were the side lengths of the smaller prism increased to get the larger prism?

2 ✓

- b) Calculate the surface area of the smaller prism.

$$SA = (4 \times 2) \times 2 + (4 \times 3) \times 2 + (2 \times 3) \times 2$$

$$= 52 \quad \checkmark$$

- c) Calculate the surface area of the larger prism.

$$SA = (8 \times 6) \times 2 + (8 \times 4) \times 2 + (4 \times 6) \times 2$$

$$SA = 208 \quad \checkmark$$

- d) Calculate $\frac{\text{Surface area of large prism}}{\text{Surface area of small prism}}$ $\frac{208}{52} = 4$ ✓

- e) Explain the relationship between the answers to parts (a) and (d).

If a side has increased by a scale factor of 2,
surface area will be increased by a scale
factor of 4.

✓

12. [1 mark]

Complete the following sentence:

If the side lengths of any figure are increased by a scale factor of n , then the surface area of similar figures are increased by a scale factor of n^2 .

13. [3 marks]

A photograph and its enlargement are similar in shape. The smaller photograph has an area of 56 cm^2 , and the larger photograph has an area of 84 cm^2 . If the larger photograph has a length of 18 cm, what is the length of the smaller frame?

$$\frac{84}{56} = 1.5 \checkmark$$

$$\sqrt{1.5} = 1.2247 \checkmark$$

$$18 \div 1.2247 = 14.7 \checkmark$$